Coding / Final Experimental Protocol

PDF edition of preserved repository document | docs/coding/final_experimental_protocol.md

Historical statements and test counts below retain their original scope. Consult the master report for current status.

Frozen experimental protocol

Preserve original protocols, results and multiplication negative findings. Recover useful jobs and checkpoints; distinguish teacher collection, retrieval adaptation and gradient training. The recovery inventory records deliberately deferred retrieval inference and its retained 2,400-example accepted prefix.

The frozen generator dataset contains 5,000 training, 20 validation, 20 IID test and 20 structural OOD tasks. Ten training categories cover simple Python bugs; four OOD categories hold out function interaction, cross-file repair, compound bugs and unfamiliar filtering structure. Every correct/buggy pair is actually executed. IID shares semantic templates with training intentionally; foundation-model contamination remains unknown. Learning pools have prior diagnostic exposure, disclosed explicitly.

The independent final lockbox contains 100 fresh tasks: 50 IID and 50 OOD, seed 13001, excluding every prior training/validation/evaluation seed. Its SHA-256 is 392ef68a412f8df49b93af60ee24318a6746a78aaeee30acdc0d6f72e92bc8a5. The final configuration is fixed at training budget 100, seed 11, three edits/eight actions, before lockbox results. Evaluate each condition once; no lockbox-driven repair, tuning or selection. Deterministic resumability reuses completed episodes and does not resample unsuccessful tasks.

The gradient comparison measures budgets 10, 25, 50 and 100 with three independent adapter initialization seeds 11,23,37. Plain-patch training additionally retains seeds 53,71. Nested checkpoints follow identical ordered task prefixes and one training epoch; examples are unique tasks, not optimizer steps, tokens or retrieved demonstrations. A zero-domain-training control supplies N=0. Requested larger gradient budgets are unexecuted and cannot be inferred from collecting 5,000 trajectories. The practical compute limit is documented using actual observed training time/memory. This reduced range right-censors unresolved thresholds.

Accuracy uncertainty averages observed seed outcomes within each task and bootstraps paired task clusters, 2,000 draws, fixed analysis seed17. Report observed per-seed means and sample SD separately. These task-cluster intervals do not establish a population-wide training-seed interval. N* is the minimum observed budget reaching 80/90/95 percent, without interpolation. Unreached thresholds are right censored; DER is undefined for missing thresholds or zero denominator. Normalized trapezoidal AULC integrates accuracy over log positive budgets; N=0 is reported separately. Paired deltas compare identical task keys. Public pass@k is never inferred from interacting edits.

Actual post-termination snapshot verification supplies interactive repair-proposal success@1/3/5 in the recovery diagnostic. Invalid unparseable proposals count as failed attempts; inspection/testing actions are excluded. First-applied-patch success in the final study is separately labeled and is not single-call capability. Syntax, incorrect-return, regression and nontermination faults are executed visibly. No private verdict is fed

back during policy actions. The final study allows at most three edits; success@5 is unavailable there. Invalid parsed actions, invalid executed transitions, terminal goal and grounded success are distinct quantities.

Inference ablations remove state, relations, memory, explicit goal, exemplar trajectory, environmental execution feedback or hypothetical planning while keeping trained weights fixed. They do not remove latent knowledge learned during training. Robustness perturbations rename functions/variables, reorder files/tests, add distractors, relocate packages and reformat source. Each perturbation's oracle and defect are actually checked; unchanged/inapplicable variants are omitted. Small diagnostic samples do not substantiate universal robustness.

Latency runs disable generation caching and run after other model experiments. Report per-episode inclusive stage times, fresh token-generation TTFT, wall time and model/tool/token counts. Stage timings are nested, so cannot be summed as exclusive components. Fourteen cases per condition make P99 exploratory. Ordinary OS activity is uncontrolled. Cached research episodes remain valid accuracy records but cannot support fresh performance claims.

Public HumanEval uses all164 official pinned tasks, one greedy completion, max512 new tokens, full-module instruction prompts and actual WASI execution. It has no adaptation, few-shot examples or feedback. Report pass@1 only. Preserve the earlier 0.5B result; compare the selected1.5B model under the same local protocol. This prompt/runtime differs from historical API reports, whose exact sampling settings may be undisclosed. No general GPT/Gemini superiority claim follows.

Robustness/ablation runs also disable generation caching, allowing measured dream/memory timing deltas. A supplemental no-trajectory-supervision diagnostic switches to the patch-only adapter at the same100-task budget/seed and keeps the typed action loop. Training input/label changes make this a combined intervention rather than isolated removal. The explicit-goal ablation removes policy goal text; terminal private assessment remains necessary to measure success. A “no terminal goal evaluator” result would be unmeasurable and is not claimed.

TTFT measures the first decoder token after generation starts, excluding model load and preceding prompt tokenization. Peak RSS is the host process high-water mark; fresh latency shares one process across sequential configurations, so its per-row RSS is not an isolated per-task peak. WASI guest limits are separately recorded. Test-suite executions count actual VM calls, including terminal assessment, rather than claiming every expected case ran when import/compile fails.